Linear Regression and OLS

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Linear Regression & OLS Practice Exercises**

Solve the initial value problem for the coupled system of linear differential equations:

$\mathbf{x}'(t) = (1, 4; 2, 3)\mathbf{x}(t), \qquad \mathbf{x}(0) = (5; 1).$

Step 1: Eigenvalues of coefficient matrix: $\det(A - \lambda I) = (1 - \lambda)(3 - \lambda) - 8 = \lambda^2 - 4\lambda - 5 = (\lambda - 5)(\lambda + 1) = 0 \Rightarrow \lambda_1 = 5,\ \lambda_2 = -1.$ Step 2: Eigenvectors:For $\lambda_1 = 5$: $(A - 5I) = (-4, 4; 2, -2) \Rightarrow \mathbf{v}_1 = (1; 1)$.For $\lambda_2 = -1$: $(A + I) = (2, 4; 2, 4) \Rightarrow \mathbf{v}_2 = (-2; 1)$. Step 3: General solution and initial conditions.

2 **[Foundational]** • **Target: Linear Regression & OLS Practice Exercises**

Level 3 · Advanced / University RigorFind the general solution in parametric vector form for the underdetermined linear system: $x_1 + 2x_2 - 3x_3 + x_4 + 2x_5, = 3;\ -x_1 - 2x_2 + 4x_3 - 2x_4 - x_5, = -2;\ 2x_1 + 4x_2 - 5x_3 + x_4 + 5x_5, = 7$

Step 1 · Set up Augmented Matrix and Eliminate: x Perform $R_3 \leftarrow R_3 - R_2$ and $R_1 \leftarrow R_1 + 3R_2$ to reach RREF: $(1, 2, 0, -2, 5, |, 6; 0, 0, 1, -1, 1, |, 1; 0, 0, 0, 0, 0, |, 0).$ Step 2 · Identify Basic and Free Variables: Pivots in columns 1 and 3 $\Rightarrow x_1, x_3$ are basic. Columns 2, 4, 5 lack pivots $\Rightarrow x_2 = r, x_4 = s, x_5 = t$ are free parameters. Step 3 · Express Basic Variables: $x_1 = 6 - 2r + 2s - 5t, \qquad x_3 = 1 + s - t.$ Step 4 · Parametric Vector.

3 **[Foundational]** • **Target: Linear Regression & OLS Practice Exercises**

Level 3 · AdvancedA scatterplot displays a point at $x = 18$, where the rest of the sample data has x-values between 2 and 8. The point falls almost exactly along the linear trend established by the other points. Explain the effect of this point on: (a) leverage, (b) the residual, (c) the correlation coefficient r, and (d) the slope b_1 of the LSRL.

(a) Leverage: The point has high leverage because its x-value (18) is substantially far from the mean $\bar{x}$ of the remaining data. (b) Residual: The point has a very small residual because it falls directly on the linear trajectory established by the other data points. (c) Correlation r: Because the point has an extreme x -value and small residual in line with the trend, it substantially strengthens (increases) the correlation coefficient r toward 1. (d) Slope b_1: The point is not influential on the s.

4 **[Foundational]** • **Target: Linear Regression & OLS Practice Exercises**

Find the standard matrix $[T]$ of the linear operator $T : \mathbb{R}^2 \to \mathbb{R}^2$ that performs a horizontal shear by factor 2, followed by a counterclockwise rotation of 45°.

Step 1: Standard matrix of shear $[S]$: $S(x, y) = (x + 2y, y) \Rightarrow [S] = (1, 2; 0, 1).$ Step 2: Standard matrix of rotation $[R]$: $[R] = (\cos(45^\circ), -\sin(45^\circ); \sin(45^\circ), \cos(45^\circ)) = \frac{\sqrt{2}}{2}(1, -1; 1, 1).$ Step 3: Compose operators $[T] = [R][S]$.

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5 [Foundational] • Target: Linear Regression & OLS Practice Exercises

Level 3 · Advanced / University RigorLet $A = (2, 1; 0, 3)$. Using the Cayley–Hamilton Theorem, express A^4 and A^{-1} as linear combinations $c_1A + c_0I$.

Step 1 · Find Characteristic Polynomial: $p(\lambda) = (2-\lambda)(3-\lambda) - 0 = \lambda^2 - 5\lambda + 6$. Step 2 · Apply Cayley–Hamilton: $A^2 - 5A + 6I = 0$: From this: $A^2 = 5A - 6I$. Multiply by A: $A^3 = 5A^2 - 6A = 5(5A - 6I) - 6A = 25A - 30I - 6A = 19A - 30I$. Multiply by A again: $A^4 = 19A^2 - 30A = 19(5A - 6I) - 30A = 95A - 114I - 30A = \mathbf{65A - 114I}$. Step 3 · Express A^{-1}: Multiply $A^2 - 5A + 6I = 0$ by A^{-1}.

6 [Foundational] • Target: Linear Regression & OLS Practice Exercises

Using the regression model $\hat{y} = 38.0 + 0.55x$, a student who completed 80% of homework scored 86 on the exam. (a) Calculate the residual. (b) Explain what a random scatter of points around 0 on a residual plot indicates about model adequacy.

(a) Residual: Predicted score: $\hat{y} = 38.0 + 0.55(80) = 38.0 + 44.0 = 82.0$. Residual: $e = y - \hat{y} = 86 - 82.0 = \mathbf{+4.0}$ points. The student scored 4.0 points higher than predicted by the regression line. (b) Residual Plot Adequacy: A residual plot showing random scatter with no discernible curve, fanning, or systematic pattern confirms that a linear model is appropriate and that error variance is approximately constant.

7 [Foundational] • Target: Linear Regression & OLS Practice Exercises

Level 2 · MediumFind all solutions to the homogeneous linear system:
$x_1 + 3x_2 - 2x_3, = 0; 2x_1 + 6x_2 - 4x_3, = 0$

Step 1: Set up and row reduce the coefficient matrix: $(1, 3, -2; 2, 6, -4) \to (1, 3, -2; 0, 0, 0)$. Step 2: Pivot is in column 1 (x_1 basic); columns 2 and 3 have no pivots ($x_2 = s, x_3 = t$ free). Step 3: Solve for $x_1 = -3s + 2t$. Write in parametric vector form: $\mathbf{x} = (x_1; x_2; x_3) = \mathbf{s}(\mathbf{-3; 1; 0}) + \mathbf{t}(\mathbf{2; 0; 1}), \quad \mathbf{s, t} \in \mathbb{R}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

A least-squares regression line predicting exam score (y) from homework completion percentage (x) is $\hat{y} = 38.0 + 0.55x$ with correlation $r = 0.82$. (a) Interpret the slope 0.55 in context. (b) Calculate and interpret the coefficient of determination r^2.

(a) Slope Interpretation: For each additional 1 percentage point increase in homework completion, the model predicts an average increase of 0.55 points on the exam. (b) r^2 Calculation & Interpretation: $r^2 = (0.82)^2 = 0.6724 = \mathbf{67.24\%}$. Exactly 67.24% of the variability in exam scores is explained by the linear relationship with homework completion percentage.

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9 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

Level 1 · FoundationsConvert the matrix into Reduced Row Echelon Form (RREF): $M = (1, -2, 1; 2, -4, 5)$.

Step 1 · Eliminate below pivot 1: Perform $R_2 \leftarrow R_2 - 2R_1$: $(1, -2, 1; 0, 0, 3)$. Step 2 · Scale pivot 2 to 1: Perform $R_2 \leftarrow \frac{1}{3}R_2$: $(1, -2, 1; 0, 0, 1)$. Step 3 · Eliminate above pivot 2: Perform $R_1 \leftarrow R_1 - R_2$: $(\mathbf{1, -2, 0; 0, 0, 1})$.

10 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

A linear transformation $T : \mathbb{R}^7 \to \mathbb{R}^4$ has $\dim(\ker T) = 3$. What is the dimension of the range of T, and is T surjective?

By the Dimension Theorem, $\text{nullity}(T) + \text{rank}(T) = \dim(\text{domain})$. Here: $3 + \text{rank}(T) = 7 \Rightarrow \text{rank}(T) = \dim(\text{im}T) = 4$. Since $\dim(\text{im}T) = 4 = \dim(\mathbb{R}^4)$, the range spans the entire codomain, so T is surjective (onto).

11 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

Let $f(x, y) = x^3e^{2y} - y\ln(x)$ defined on $x > 0$. Compute the first-order partial derivatives $f_x(x, y)$ and $f_y(x, y)$, and evaluate $f_x(1, 0)$ and $f_y(1, 0)$.

1. Partial Derivative with respect to x (treating y as constant): $f_x(x, y) = \frac{\partial}{\partial x}[x^3e^{2y}] - \frac{\partial}{\partial x}[y\ln x] = 3x^2e^{2y} - \frac{y}{x}$. At $(1, 0)$: $f_x(1, 0) = 3(1)^2e^0 - \frac{0}{1} = 3$. 2. Partial Derivative with respect to y (treating x as constant): $f_y(x, y) = \frac{\partial}{\partial y}[x^3e^{2y}] - \frac{\partial}{\partial y}[y\ln x] = 2x^3e^{2y} - \ln(x)$. At $(1, 0)$: $f_y(1, 0) = 2(1)^3e^0 - \ln(1) = 2 - 0 = 2$.

12 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

Classify the critical points $(0, 0)$ and $(1, 1)$ of $f(x, y) = x^3 + y^3 - 3xy$ using the Second Derivative Test and Hessian matrix.

1. Compute the second partial derivatives: $f_{xx} = 6x, \quad f_{yy} = 6y, \quad f_{xy} = -3$. The Hessian determinant discriminant is: $D(x, y) = f_{xx}f_{yy} - (f_{xy})^2 = (6x)(6y) - (-3)^2 = 36xy - 9$. 2. Evaluate at $(0, 0)$: $D(0, 0) = 36(0) - 9 = -9 < 0$. Since $D < 0$, $(0, 0)$ is a saddle point. 3. Evaluate at $(1, 1)$: $D(1, 1) = 36(1)(1) - 9 = 27 > 0, \quad f_{xx}(1, 1) = 6(1) = 6 > 0$. Since $D > 0$ and $f_{xx} > 0$, $(1, 1)$ is a local minimum with value $f(1, 1) = 1 + 1 - 3 = -1$.

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13 **[Intermediate]** • **Target: Linear Regression & OLS Practice Exercises**

Use Stokes' Theorem to evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = \langle -y^3, x^3, -z^3 \rangle$ and C is the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $z = 1$, oriented counterclockwise when viewed from above. Verify surface independence by comparing the flux of $abla \times \mathbf{F}$ through the flat disk $S_1 : x^2 + y^2 \le 1, z = 1$ and the paraboloid cap $S_2 : z = 2 - (x^2 + y^2)$.

1. Compute $abla \times \mathbf{F}$: $abla \times \mathbf{F} = \left|\mathbf{i}, \mathbf{j}, \mathbf{k}; \partial_x, \partial_y, \partial_z; -y^3, x^3, -z^3\right| = \langle 0 - 0,\ 0 - 0,\ 3x^2 - (-3y^2) \rangle = \langle 0,\ 0,\ 3(x^2 + y^2) \rangle$. 2. Flux through flat disk S_1 ($z = 1$, $\mathbf{n} = \mathbf{k}$): On S_1, $d\mathbf{S} = \mathbf{k}\, dA$. $\iint_{S_1} (abla \times \mathbf{F}) \cdot d\mathbf{S} = \iint_D 3(x^2 + y^2)\, dA$. In polar coordinates.

14 **[Intermediate]** • **Target: Linear Regression & OLS Practice Exercises**

Find all eigenvalues, their algebraic and geometric multiplicities, and a basis for each eigenspace for: $A = (3, 1, 0; 0, 3, 0; 0, 0, 2)$. Is A diagonalizable?

Step 1: Characteristic equation: Since A is upper triangular, its eigenvalues are the diagonal entries: $\det(A - \lambda I) = (3 - \lambda)^2 (2 - \lambda) = 0$. Distinct eigenvalues: $\lambda_1 = 3$ (algebraic multiplicity $\alpha_1 = 2$), and $\lambda_2 = 2$ (algebraic multiplicity $\alpha_2 = 1$). Step 2: Eigenspace for $\lambda = 3$: $(A - 3I)\mathbf{v} = (0, 1, 0; 0, 0, 0; 0, 0, -1)(v_1; v_2; v_3) = (0; 0; 0)$. We obtain $v_2 = 0$ and $-v_3 = 0 \Rightarrow v_3 = 0$, w.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Linear Regression & OLS Practice Exercises** • **Cross-Lesson: 7.01 Logistic Regression and Classification**

Apply the Gram–Schmidt orthogonalization process to the columns of A to find its QR factorization: $A = (1, 2; 2, 1; -2, 2)$.

Step 1: Orthogonalize $\mathbf{x}_1 = (1, 2, -2)^T$ and $\mathbf{x}_2 = (2, 1, 2)^T$: First vector: $\mathbf{v}_1 = \mathbf{x}_1 = (1; 2; -2)$. Norm: $||\mathbf{v}_1|| = \sqrt{1^2 + 2^2 + (-2)^2} = \sqrt{9} = 3$. Subtract projection of $\mathbf{x}_2$ onto $\mathbf{v}_1$: $\mathbf{x}_2 \cdot \mathbf{v}_1 = 2(1) + 1(2) + 2(-2) = 2 + 2 - 4 = 0$. Since $\mathbf{x}_2 \cdot \mathbf{v}_1 = 0$, the columns were already orthogonal! Thus $\mathbf{v}_2 = \mathbf{x}_2 = (2; 1; 2)$. Norm.

16 **[Challenging]** • **Target: Linear Regression & OLS Practice Exercises** • **Cross-Lesson: 8.01 PCA and Dimensionality Reduction**

Use the Divergence Theorem to calculate the outward flux of $\mathbf{F} = \langle x^3 + y\sin z,\ y^3 + z\cos x,\ z^3 + xy \rangle$ across the closed boundary surface of the solid cylinder $E : x^2 + y^2 \le a^2,\ 0 \le z \le h$.

1. Compute the divergence $abla \cdot \mathbf{F}$: $abla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x^3 + y\sin z) + \frac{\partial}{\partial y}(y^3 + z\cos x) + \frac{\partial}{\partial z}(z^3 + xy) = 3x^2 + 3y^2 + 3z^2 = 3(x^2 + y^2 + z^2)$. Notice that all cross-coupling transcendental terms vanish identically. 2. Apply the Divergence Theorem: $\iint_{\partial E} \mathbf{F} \cdot d\mathbf{S} = \iiint_E 3(x^2 + y^2 + z^2)\, dV$. 3. Convert to cylindrical coordinates: In E, $x^2 + y^2 = r^2$, $0 \le r \le a$.

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17 **[Challenging]** • **Target: Linear Regression & OLS Practice Exercises** • **Cross-Lesson: 1.01 Vectors and Embeddings in AI**

Prove that if λ is an eigenvalue of an invertible matrix A, then $\lambda eq 0$ and λ^{-1} is an eigenvalue of A^{-1} with the exact same corresponding eigenvector.

Step 1: Show $\lambda eq 0$: By definition of an eigenvalue, there exists a nonzero vector $\mathbf{v} eq \mathbf{0}$ such that $A\mathbf{v} = \lambda\mathbf{v}$. If $\lambda = 0$, then $A\mathbf{v} = \mathbf{0}$, which means $\mathbf{v} \in \text{Null}(A)$, contradicting the assumption that A is invertible. Thus $\lambda eq 0$. Step 2: Show λ^{-1} is an eigenvalue of A^{-1}: Multiply both sides of $A\mathbf{v} = \lambda\mathbf{v}$ by A^{-1} from the left.

18 **[Challenging]** • **Target: Linear Regression & OLS Practice Exercises** • **Cross-Lesson: 2.01 Neural Layer Matrices and Activations**

Prove that the 3D vector field $\mathbf{F} = \langle 2xy + z^2,\ x^2 + 2yz,\ 2xz + y^2 \rangle$ is conservative on $\mathbb{R}^3$, construct a potential function f satisfying $abla f = \mathbf{F}$, and evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(0, 0, 0)$ to $(1, 2, 3)$.

1. Verify $abla \times \mathbf{F} = \mathbf{0}$: $abla \times \mathbf{F} = \left|\mathbf{i}, \mathbf{j}, \mathbf{k};\ \partial_x, \partial_y, \partial_z;\ 2xy + z^2, x^2 + 2yz, 2xz + y^2\right| = \langle 2y - 2y,\ 2z - 2z,\ 2x - 2x \rangle = \mathbf{0}$. Since $\mathbb{R}^3$ is simply connected, $\mathbf{F}$ is conservative. 2. Reconstruct the potential function:Integrate $f_x = 2xy + z^2 \Rightarrow f(x, y, z) = x^2y + xz^2 + g(y, z)$.Differentiate with respect to y.

19 **[Challenging]** • **Target: Linear Regression & OLS Practice Exercises** • **Cross-Lesson: 3.01 Gradient Descent and Optimization**

Diagonalize the matrix $A = (1, 2; 2, 1)$ by finding invertible P and diagonal D such that $A = PDP^{-1}$. Then compute A^k for any positive integer k.

Step 1: Eigenvalues of A: $\det(A - \lambda I) = (1 - \lambda)^2 - 4 = \lambda^2 - 2\lambda - 3 = (\lambda - 3)(\lambda + 1) = 0 \Rightarrow \lambda_1 = 3,\ \lambda_2 = -1.$ Step 2: Eigenvectors:For $\lambda_1 = 3$: $(-2, 2; 2, -2)\mathbf{v}_1 = \mathbf{0} \Rightarrow \mathbf{v}_1 = (1; 1)$.For $\lambda_2 = -1$: $(2, 2; 2, 2)\mathbf{v}_2 = \mathbf{0} \Rightarrow \mathbf{v}_2 = (-1; 1)$. Step 3: Construct P, D, and P^{-1}.

20 **[Challenging]** • **Target: Linear Regression & OLS Practice Exercises** • **Cross-Lesson: 4.01 Conditional Probability and Bayes Rule**

Level 3 · AdvancedFind the coordinates of all points on the curve $x^2 - xy + y^2 = 9$ where the tangent line is horizontal.

Step 1 · Implicit Differentiation: Differentiate with respect to x: $2x - \left(1 \cdot y + x\frac{dy}{dx}\right) + 2y\frac{dy}{dx} = 0 \Rightarrow 2x - y + (2y - x)\frac{dy}{dx} = 0.$ Solve for $\frac{dy}{dx}$: $\frac{dy}{dx} = \frac{y - 2x}{2y - x}.$ Step 2 · Condition for Horizontal Tangent: Set the numerator equal to zero while ensuring the denominator is non-zero: $y - 2x = 0 \Rightarrow y = 2x.$ Step 3 · Substitute into Original Curve.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Linear Regression & OLS Practice Exercises

A biostatistician is investigating the relationship between the dosage of a new experimental drug (measured in milligrams) and the reduction in systolic blood pressure (in mmHg) for patients with mild hypertension. They collected data from $n = 200$ patients and fit a simple linear regression model. The model's output yielded an estimated intercept of $\hat{\beta}_0 = 5.2$ and an estimated slope of $\hat{\beta}_1 = -0.8$. The standard error of the estimated slope is $\text{SE}(\hat{\beta}_1) = 0.15$. A 95% confidence interval for the true slope β_1 was calculated to be $[-1.12, -0.48]$. The biostatistician then constructs a hypothesis test to determine if the true slope is significantly different from zero. To do this, they calculate a test statistic. If the same α level used for the confidence interval were applied to a two-tailed hypothesis test of $H_0 : \beta_1 = 0$ versus $H_a : \beta_1 eq 0$, what would be the approximate p-value associated with this test statistic?

a. Greater than 0.05

b. Equal to 0.05

c. **Less than 0.05** ✓

d. Cannot be determined from the given information

The problem asks for the approximate p-value of a two-tailed hypothesis test for the slope parameter β_1 in a simple linear regression model. We are given a 95% confidence interval for β_1 as $[-1.12, -0.48]$. A confidence interval and a hypothesis test are related. For a $100(1-\alpha)\%$ confidence interval, if the null hypothesis value (in this case, $\beta_1 = 0$) is *outside* the confidence interval, we reject H_0 at the α significance level. Conversely, if the null hypothesis value is *in.

22 [Foundational] • Target: Linear Regression & OLS Practice Exercises

Consider a linear transformation represented by the matrix $A = (3, 0; 0, -2)$. If $\mathbf{v} = (1; 1)$ is an eigenvector of A, what is the corresponding eigenvalue?

a. **3** ✓

b. -2

c. 1

d. 5

An eigenvector $\mathbf{v}$ of a matrix A satisfies the equation $A\mathbf{v} = \lambda\mathbf{v}$, where λ is the corresponding eigenvalue. We are given the matrix $A = (3, 0; 0, -2)$ and an eigenvector $\mathbf{v} = (1; 1)$. Let's compute $A\mathbf{v}$:
$A\mathbf{v} = (3, 0; 0, -2)(1; 1) = ((3)(1) + (0)(1); (0)(1) + (-2)(1)) = (3; -2)$ Now, we need to find λ such that $A\mathbf{v} = \lambda\mathbf{v}$, which means.

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23 [Foundational] • Target: Linear Regression & OLS Practice Exercises

Consider a linear transformation represented by the matrix $A = (2, 1; 1, -2)$. A point P is mapped to P' by this transformation. If the coordinates of P are $(x; y)$ and the coordinates of P' are $(x'; y')$, then $(x'; y') = A(x; y)$. Suppose a curve is defined by the equation $x'^2 + y'^2 = 5$. What is the equation of the original curve $x^2 + y^2 = k$ in terms of x and y that transforms into this circle? Find the value of k.

a. 1

b. **2** ✓

c. 3

d. 4

The transformation is given by $(x'; y') = (2, 1; 1, -2)(x; y)$. We can express $(x; y)$ in terms of $(x'; y')$ by finding the inverse of matrix A. The determinant of A is $\det(A) = (2)(-2) - (1)(1) = -4 - 1 = -5$. The inverse matrix is $A^{-1} = \frac{1}{-5}(-2, -1; -1, 2) = (2/5, 1/5; 1/5, -2/5)$. Therefore, $(x; y) = A^{-1}(x'; y') = (2/5, 1/5; 1/5, -2/5)(x'; y')$. This giv.

24 [Foundational] • Target: Linear Regression & OLS Practice Exercises

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation represented by the matrix $A = (3, -1\ 1, 2)$ with respect to the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$. Consider a new basis $B = \{\mathbf{v}_1, \mathbf{v}_2\}$ where $\mathbf{v}_1 = (1\ 1)$ and $\mathbf{v}_2 = (1\ -1)$. Let P be the change-of-basis matrix from basis B to the standard basis. Find the matrix representation of T with respect to basis B, denoted by $[T]_B$. This matrix can be expressed as $P^{-1}AP$. Find the sum of the diagonal entries of $[T]_B$.

a. **5** ✓

b. 6

c. 4

d. 7

The change-of-basis matrix P from basis B to the standard basis has the vectors of B as its columns. Thus, $P = (1, 1\ 1, -1)$. The inverse of P is $P^{-1} = \frac{1}{\det(P)}(-1, -1\ -1, 1)$. Since $\det(P) = (1)(-1) - (1)(1) = -2$, we have $P^{-1} = \frac{1}{-2}(-1, -1\ -1, 1) = (1/2, 1/2\ 1/2, -1/2)$. The matrix representation of T with respect to basis B is given by $[T]_B = P^{-1}AP$. First, we compute AP.

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25 [Foundational] • Target: Linear Regression & OLS Practice Exercises

Consider a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^2$ defined by $T(x, y, z) = (2x - y + z, x + 3y - 2z)$. Let V be the image of T, i.e., $V = \text{Im}(T)$. Furthermore, let W be the subspace of $\mathbb{R}^2$ defined by $W = \{(a, b) \in \mathbb{R}^2 \mid a + b = 0\}$. Find the dimension of the intersection of V and W, i.e., $\dim(V \cap W)$.

a. 0

b. **1** ✓

c. 2

d. Cannot be determined

The linear transformation T maps $\mathbb{R}^3$ to $\mathbb{R}^2$. The matrix representation of T is $A = (2, -1, 1\ 1, 3, -2)$. The image of T, denoted as $V = \text{Im}(T)$, is the column space of A. The columns of A are $(2\ 1)$, $(-1\ 3)$, and $(1\ -2)$. Since these vectors span V, we can find a basis for V by finding the pivot columns of A or by noting that the image is a subspace of $\mathbb{R}^2$. The rank of A is 2, which means.

26 [Foundational] • Target: Linear Regression & OLS Practice Exercises

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation such that $T(x, y) = (3x - y, x + 2y)$. Consider the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ for $\mathbb{R}^2$, where $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$. Let $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ be another basis for $\mathbb{R}^2$, where $\mathbf{v}_1 = (1, 1)$ and $\mathbf{v}_2 = (1, -1)$. If the matrix representation of T with respect to the basis $\mathcal{B}$ is $[T]_{\mathcal{B}} = (a, b\ c, d)$, and $a + d = 5$, determine the value of $\det(T)$.

a. **7** ✓

b. 5

c. 3

d. 2

The matrix representation of T with respect to the standard basis $\mathcal{E} = \{\mathbf{e}_1, \mathbf{e}_2\}$ is $A = (3, -1\ 1, 2)$. The determinant of T is the determinant of its matrix representation with respect to any basis, so $\det(T) = \det(A) = (3)(2) - (-1)(1) = 6 + 1 = 7$. The information about the basis $\mathcal{B}$ and $[T]_{\mathcal{B}} = (a, b\ c, d)$ with $a + d = 5$ is a distractor. The trace of a matrix is invariant under a change of basis, so.

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27 [Foundational] • Target: Linear Regression & OLS Practice Exercises

A social scientist is investigating the relationship between the number of hours spent studying per week (X) and the final exam score (Y) for a group of $n = 100$ students. The least-squares regression line is given by $\hat{y} = 55.2 + 8.5x$. The standard error of the slope is $SE_b = 1.2$. The researcher wants to construct a 95% confidence interval for the true slope of the regression line. What is the value of the t-score used for this confidence interval?

a. 1.645

b. 1.960

c. **1.984 ✓**

d. 2.576

The problem asks for the t-score used to construct a 95% confidence interval for the true slope of a regression line. The degrees of freedom for a simple linear regression are $df = n - 2$, where n is the number of data points. In this case, $n = 100$, so $df = 100 - 2 = 98$. We need to find the t-score such that 95% of the area under the t-distribution curve with 98 degrees of freedom is between $-t$ and t. This means that 2.5% of the area is in each tail ($\alpha/2 = 0.025$). Looking up the t-table.

28 [Foundational] • Target: Linear Regression & OLS Practice Exercises

A solid object in $\mathbb{R}^3$ is defined by the region bounded by the planes $x = 0$, $y = 0$, $z = 0$, and the plane passing through the points $P = (6, 0, 0)$, $Q = (0, 4, 0)$, and $R = (0, 0, 3)$. If a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ is represented by the matrix $M = (1, 0, 0; 0, 1, 0; 0, 0, 2)$, what is the volume of the transformed solid object?

a. 12

b. **24 ✓**

c. 36

d. 48

The volume of the solid region bounded by the coordinate planes and the plane passing through $P = (6, 0, 0)$, $Q = (0, 4, 0)$, and $R = (0, 0, 3)$ can be calculated. The equation of the plane passing through these points is given by $\frac{x}{6} + \frac{y}{4} + \frac{z}{3} = 1$. This is a tetrahedron with vertices at $(0, 0, 0)$, $(6, 0, 0)$, $(0, 4, 0)$, and $(0, 0, 3)$. The volume of such a tetrahedron is $V_{original} = \frac{1}{6}|abc|$, where a, b, c are the intercepts on the axes. Thus.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation represented by the matrix $A = (2, k\ 1, 3)$ with respect to the standard basis. The image of the vector $\mathbf{v} = (1\ 1)$ under T is $T(\mathbf{v}) = (3\ 4)$. Find the value of the parameter k.

a. **1** ✓

b. 2

c. 3

d. 4

The linear transformation T is represented by the matrix $A = (2, k\ 1, 3)$. The action of T on a vector $\mathbf{v}$ is given by $T(\mathbf{v}) = A\mathbf{v}$. We are given that $\mathbf{v} = (1\ 1)$ and $T(\mathbf{v}) = (3\ 4)$. Therefore, we have the matrix equation: $(2, k\ 1, 3)(1\ 1) = (3\ 4)$ Performing the matrix multiplication on the left side gives: $(2(1) + k(1)\ 1(1) + 3(1)) = (2 + k\ 4)$ Equating this to.

30 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

Consider the linear transformation $T: \mathbb{R}^3 \to \mathbb{R}^3$ defined by $T(\mathbf{x}) = A\mathbf{x}$, where $A = (1, 2, 3; 4, 5, 6; 7, 8, 9)$. Let V be the image of T, i.e., $V = \text{Im}(T)$. V is a subspace of $\mathbb{R}^3$. Find the dimension of the orthogonal complement of V, denoted by $V^\perp$.

a. **1** ✓

b. 2

c. 0

d. 3

The image of the linear transformation T is the column space of the matrix A, i.e., $V = \text{Col}(A)$. The dimension of the image of T is the rank of the matrix A, i.e., $\dim(V) = \text{rank}(A)$. We can find the rank of A by reducing it to row-echelon form: $A = (1, 2, 3; 4, 5, 6; 7, 8, 9) \to (1, 2, 3; 0, -3, -6; 0, -6, -12) \to (1, 2, 3; 0, -3, -6; 0, 0, 0)$ There are two non-zero rows, so the rank of A is 2. Thus, $\dim(V) = 2$. ; ; We kn.

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31 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

Consider the following system of linear equations represented by the augmented matrix: $(1, 2, |, 5; 2, -1, |, 0)$. Use row reduction to transform the matrix into row-echelon form and determine the unique solution (x, y) for the system. What is the value of x?

a. **1** ✓

b. 2

c. 3

d. 4

We are given the augmented matrix: $(1, 2, |, 5; 2, -1, |, 0)$ To transform this matrix into row-echelon form, we can perform the following row operation: $R_2 \leftarrow R_2 - 2R_1$.
$(1, 2, |, 5; 2 - 2(1), -1 - 2(2), |, 0 - 2(5))$ $(1, 2, |, 5; 0, -5, |, -10)$ Now, we can perform another row operation: $R_2 \leftarrow R_2/-5$. $(1, 2, |, 5; 0, 1, |, 2)$ This matrix is now in row-echelon form. The corresponding system of equations is: $x + 2y = 5$ $y = 2$ Substitut.

32 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

A linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $T(x, y) = (ax + by, cx + dy)$. The matrix representation of T with respect to the standard basis is $A = (a, b\ c, d)$. The image of the standard basis vectors $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$ under T are given by $T(\mathbf{e}_1) = (3, -1)$ and $T(\mathbf{e}_2) = (1, 2)$. The transformation is applied to a square with vertices $(0, 0), (1, 0), (0, 1), (1, 1)$. The area of the transformed square is known to be k. Find the value of k.

a. **5** ✓

b. 10

c. 15

d. 20

The matrix representation of the linear transformation T with respect to the standard basis is formed by the images of the standard basis vectors as its columns. Therefore, $A = (3, 1\ -1, 2)$. The area of the transformed region is the absolute value of the determinant of the matrix A multiplied by the area of the original region. The original region is a square with vertices $(0, 0), (1, 0), (0, 1), (1, 1)$, which has an area of $1 \times 1 = 1$. The determinant of A is.

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33 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

Consider the linear transformation $T: \mathbb{R}^2 \to \mathbb{R}^2$ represented by the matrix $A = (3, 1; 1, 3)$. Identify one of the eigenvalues of A.

a. 1

b. 2

c. 3

d. **4** ✓

To find the eigenvalues of a matrix A, we solve the characteristic equation $\det(A - \lambda I) = 0$, where λ represents the eigenvalues and I is the identity matrix. For the given matrix $A = (3, 1; 1, 3)$, the characteristic equation is: $\det(3 - \lambda, 1; 1, 3 - \lambda) = 0$ This expands to: $(3 - \lambda)(3 - \lambda) - (1)(1) = 0$ $(3 - \lambda)^2 - 1 = 0$ $9 - 6\lambda + \lambda^2 - 1 = 0$ $\lambda^2 - 6\lambda + 8 = 0$ Factoring the quadratic equation.

34 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

Consider the linear transformation $T: \mathbb{R}^2 \to \mathbb{R}^2$ represented by the matrix $A = (1, 2; k, 4)$. The image of the vector $\mathbf{v} = (1; 1)$ under this transformation is $T(\mathbf{v}) = (3; 5)$. Determine the value of the parameter k.

a. **1** ✓

b. 2

c. 3

d. 4

The linear transformation T is defined by $T(\mathbf{x}) = A\mathbf{x}$. We are given $A = (1, 2; k, 4)$ and $\mathbf{v} = (1; 1)$. The image of $\mathbf{v}$ is $T(\mathbf{v}) = A\mathbf{v} = (1, 2; k, 4)(1; 1) = (1 \cdot 1 + 2 \cdot 1; k \cdot 1 + 4 \cdot 1) = (3; k + 4)$. We are given that $T(\mathbf{v}) = (3; 5)$. Equating the components, we have $3 = 3$ (which is consistent) and $k + 4 = 5$. Solving for k, we get.

35 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

Consider the linear transformation $T: \mathbb{R}^2 \to \mathbb{R}^2$ represented by the matrix $A = (2, 0\ 0, 3)$. What is the image of the vector $\mathbf{v} = (1\ 1)$ under this transformation, i.e., find $T(\mathbf{v})$?

a. (1 1)

b. (2 3) ✓

c. (3 2)

d. (4 9)

The linear transformation T is represented by the matrix A. Therefore, the image of a vector $\mathbf{v}$ under T is given by $T(\mathbf{v}) = A\mathbf{v}$. In this case, $A = (2, 0\ 0, 3)$ and $\mathbf{v} = (1\ 1)$. So, $T(\mathbf{v}) = (2, 0\ 0, 3)(1\ 1) = (2 \times 1 + 0 \times 1\ 0 \times 1 + 3 \times 1) = (2\ 3)$.

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36 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

Consider the linear transformation $T: \mathbb{R}^2 \to \mathbb{R}^2$ represented by the matrix $A = (1, 2; 3, 4)$. If a vector $\mathbf{v} = (2; 1)$ is transformed by T, what is the resulting vector $T(\mathbf{v})$?

a. $(4; 10)$ ✓

b. $(2; 3)$

c. $(4; 7)$

d. $(3; 5)$

The linear transformation T can be applied to a vector $\mathbf{v}$ by multiplying the matrix A by the vector $\mathbf{v}$. $T(\mathbf{v}) = A\mathbf{v} = (1, 2; 3, 4)(2; 1)$ To perform this matrix-vector multiplication, we take the dot product of each row of the matrix with the vector: Row 1: $(1)(2) + (2)(1) = 2 + 2 = 4$ Row 2: $(3)(2) + (4)(1) = 6 + 4 = 10$ Therefore, the resulting vector is $(4; 10)$.

37 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

The scatterplot below displays the relationship between the number of hours (h) students spent studying for a math test and their scores (s) on that test. The equation of the least-squares regression line for this data is given by $s = 1.2h + 78.5$. If a student studied for 5 hours, what is their predicted score based on this model?

a. 91.5

b. **84.5** ✓

c. 81.5

d. 90.5

The equation of the least-squares regression line is given by $s = 1.2h + 78.5$, where s represents the predicted score and h represents the number of hours spent studying. To find the predicted score for a student who studied for 5 hours, we substitute $h = 5$ into the equation: $s = 1.2(5) + 78.5$ $s = 6.0 + 78.5$ $s = 84.5$ Therefore, the predicted score for a student who studied for 5 hours is 84.5.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Linear Regression & OLS Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

A linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $T(\mathbf{x}) = A\mathbf{x}$, where $A = (2, 1; 1, 3)$. If $\mathbf{v} = (1; 2)$, find $T(\mathbf{v})$.

a. $(4; 7)$ ✓

b. $(3; 7)$

c. $(4; 5)$

d. $(5; 7)$

To find $T(\mathbf{v})$, we multiply the matrix A by the vector $\mathbf{v}$. $T(\mathbf{v}) = A\mathbf{v} = (2, 1; 1, 3)(1; 2)$. Performing the matrix-vector multiplication: $T(\mathbf{v}) = ((2 \times 1) + (1 \times 2); (1 \times 1) + (3 \times 2)) = (2 + 2; 1 + 6) = (4; 7)$. Thus, the correct answer is $(4; 7)$.

39 [Challenging] • Target: Linear Regression & OLS Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

Consider the following system of linear equations represented by an augmented matrix: $[1, 2, |, 5; 0, 1, |, 2]$. What is the value of x in the first equation?

a. 1

b. 2

c. **3** ✓

d. 4

The augmented matrix represents the system of equations: $x + 2y = 5$ and $y = 2$. We are asked to find the value of x. We can use substitution to solve for x. Substitute the value of y from the second equation into the first equation: $x + 2(2) = 5$. This simplifies to $x + 4 = 5$. Subtracting 4 from both sides gives $x = 1$. Therefore, the value of x in the first equation is 1.

40 [Challenging] • Target: Linear Regression & OLS Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

Consider the system of linear equations represented by the following augmented matrix: $(1, 2, |, 5; 0, 1, |, 2)$ Use row reduction to determine the unique solution for x and y in the system. What is the value of x?

a. 1

b. 2

c. **3** ✓

d. 4

The augmented matrix represents the system of equations: 1) $x + 2y = 5$ 2) $y = 2$ Since the second equation directly gives the value of y as 2, we can substitute this into the first equation to find x. Substitute $y = 2$ into equation (1): $x + 2(2) = 5$ $x + 4 = 5$ $x = 5 - 4$ $x = 1$ Therefore, the unique solution is $x = 1$ and $y = 2$. The question asks for the value of x.

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41 [Challenging] • Target: Linear Regression & OLS Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

The scatterplot below displays the relationship between the number of hours students spent studying for a test and their scores on that test. The equation of the least-squares regression line is approximately $\hat{y} = 55.2 + 4.8x$, where x is the number of hours spent studying and $\hat{y}$ is the predicted test score.

a. 55.2

b. **4.8** ✓

c. 50

d. 70

The slope of the least-squares regression line represents the average change in the response variable (test score) for a one-unit increase in the explanatory variable (hours studying). In the given equation $\hat{y} = 55.2 + 4.8x$, the coefficient of x is 4.8. Therefore, for each additional hour a student studies, their predicted test score increases by an average of 4.8 points.

42 [Challenging] • Target: Linear Regression & OLS Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

Which of the following is a subspace?

a. x ✓

b. $\{p \in P_2 : p(0) = 1\}$

c. x

d. $\{A \in M_{2\times 2} : \det A = 0\}$

x is a subspace: a line through the origin, closed under addition and scaling. $\{p \in P_2 : p(0) = 1\}$: the zero polynomial is not in it. x: it does not contain the zero vector. $\{A \in M_{2\times 2} : \det A = 0\}$: not closed under addition: diag(1,0)+diag(0,1)=I.

43 [Challenging] • Target: Linear Regression & OLS Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

A linear map T satisfies $T\langle -1, -1\rangle = \langle 0, -2\rangle$ and $T\langle 0, -1\rangle = \langle 2, -1\rangle$. Find $T\langle 2, -3\rangle$.

a. $\langle -6, -1\rangle$

b. $\langle -10, 1\rangle$

c. $\langle 10, -1\rangle$ ✓

d. $\langle 11, -1\rangle$

Write $\langle 2, -3\rangle = c_1\langle -1, -1\rangle + c_2\langle 0, -1\rangle$ with $(c_1, c_2) = \langle -2, 5\rangle$. By linearity, $T\langle 2, -3\rangle = c_1\langle 0, -2\rangle + c_2\langle 2, -1\rangle = \langle 10, -1\rangle$.

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44 [Challenging] • Target: Linear Regression & OLS Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

Find the image of $\langle 1, 1\rangle$ under the rotation counterclockwise by 30°.

a. $\langle -\frac{1}{2}+\frac{\sqrt{3}}{2}, \frac{1}{2}+\frac{\sqrt{3}}{2}\rangle$ ✓

b. $\langle \frac{1}{2}+\frac{\sqrt{3}}{2}, -\frac{1}{2}+\frac{\sqrt{3}}{2}\rangle$

c. $\langle \frac{1}{2}-\frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}-\frac{1}{2}\rangle$

d. $\langle 1, 1\rangle$

The standard matrix is $\left[\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}; \frac{1}{2}, \frac{\sqrt{3}}{2}\right)\right]$; multiplying gives $\langle -\frac{1}{2}+\frac{\sqrt{3}}{2}, \frac{1}{2}+\frac{\sqrt{3}}{2}\rangle$.

45 [Challenging] • Target: Linear Regression & OLS Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

Solve $y' - \frac{y}{x} = x^2$ (linear first-order):

a. $y = \frac{1}{2}x^3 + Cx$ ✓

b. $y = \frac{1}{3}x^3 + C$

c. $y = x^3 + \frac{C}{x}$

d. $y = x^3 + Cx$

The integrating factor is $\mu(x) = e^{-\int \frac{1}{x}\,dx} = \frac{1}{x}$. Then $\left(\frac{y}{x}\right)' = x \Rightarrow \frac{y}{x} = \frac{x^2}{2} + C \Rightarrow y = \frac{1}{2}x^3 + Cx$.

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